

Luxshan Thuraisingam

Software Engineer

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Summary

Software Engineer with internship experience at **PickMe**. Proven expertise in designing complex systems including a three-layer image forensics system and benchmarked distributed message brokers for peak-load passenger handling. Built ML-driven auto-scaling framework for VMs in cloud reducing energy consumption while meeting SLOs. Proficient in **Java**, **Go**, Docker, and containerized backend development.

Experience

- **Software Engineer Intern** Mar 2025 – Sep 2025
Digital Mobility Solutions Lanka PLC (**PickMe**), Colombo, Sri Lanka
 - Developed **RESTful API** endpoints for QR payment workflow and built React Native components for operational demo application.
 - Containerized a frontend application using **Docker**, pushed the image to **Artifact Registry**, and deployed it to a **Kubernetes** cluster, enabling scalable and production-ready frontend delivery.
 - Designed and implemented a three-layer image forensics system to detect tampered driver documents using Error Level Analysis (ELA), JPEG structure decoding, and EXIF metadata analysis, integrated with **React frontend**, **Golang backend**, and containerized with **Docker**.
 - Evaluated and benchmarked Golang-based distributed queue technologies across workloads ranging from 10 to 100,000 requests, analyzing throughput and latency characteristics to optimize passenger request handling during peak demand.
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Technical Skills

- **Languages & Frameworks:** Java (Spring Boot), Go, Python, React.js, React Native.
 - **Backend & Distributed Systems:** Kafka, Redis, RabbitMQ, MCP, ETCD, MySQL, REST API
 - **DevOps & Infrastructure:** Docker, Kubernetes (GKE, Minikube), Git, GitHub Actions (CI/CD)
 - **Testing & Analysis:** Serenity BDD, Kibana, SonarQube
 - **ML & Data:** PyTorch, Scikit-learn, Pandas, Dask, SciPy, Jupyter Notebook
 - **Concepts:** Data Structures & Algorithms, OOP, SOLID, Design Patterns, Microservices Architecture
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Education

- **B.Sc. (Hons) in Information Technology** (2022 – 2026)
University of Moratuwa, Colombo, Sri Lanka
CGPA - **3.55/4.0**
 - **G.C.E. Advanced Level** (2019)
J/Mahajana College, Tellippalai, Sri Lanka
Physical Science Stream, Results: 2As 1C, Z-Score: **2.0104**
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Research

- **A VM-Aware Energy-Optimal Model Predictive Control Framework for Proactive Horizontal Autoscaling of Virtual Machines** (Ongoing)
Final Year Research Project (Individual)
Technologies: Python, PyTorch, Scikit-learn, Pandas, Dask, SciPy, Jupyter Notebooks
 - Designed a VM-Aware Multi-Step MPC controller that proactively auto scales cloud VMs by integrating ML-forecasted workloads (90–120s ahead) into a receding-horizon optimizer saving (27.8%) energy vs a fixed baseline.
 - Built an Multi layer neural network surrogate model to forecast cloud energy consumption and latency based on underlying VM workload.
 - Developed an ML workload forecaster (HistGradientBoosting, 32 temporal features) predicting workload at t+90s/105s/120s with 97.9% correlation.
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Projects

1. LabourLINK (Labour Hiring System) - 2nd Year Group Project

Technologies: React.js, React Native, Spring Boot, MySQL

[GitHub](#)

Client: AlphaCodes (Pvt) Ltd

- Developed map-integrated on-demand labor hiring web and mobile apps with real-time communication.
- Contributions (Team Leader & Full-Stack Developer):
 - Implemented authentication & session management using Spring Security and JWT, OTP-based password reset with JavaMailSender.
 - Built profile, bookings, review, rating, and reporting functionalities.
 - Created performance reports with MUI charts and filtering.

2. Message Broker Benchmarking System - Personal R&D Project (Ongoing)

Technologies: Golang, Apache Kafka, Redis, RabbitMQ, Docker

[GitHub](#)

- Developing a Go-based benchmarking framework to evaluate distributed message brokers under high-concurrency workloads.
- Implementing multi-producer/multi-consumer simulations to measure throughput, latency, and reliability across Kafka, RabbitMQ, and Redis-based queues.
- Designing configurable load scenarios and automated experiments to compare performance trade-offs and scalability characteristics.

3. LinkNestSocialMedia - Individual Project

Technologies: React.js, Spring Boot, MySQL, Docker, GitHub Actions

[GitHub](#)

- Built a full-stack social media application on a microservices architecture, supporting posts, likes, and profile management.
- Implemented JWT-based authentication and secure password reset using JavaMailSender.
- Set up CI/CD pipelines with GitHub Actions, including automated JUnit testing and Dockerized.

4. Distributed Queue Evaluation - Personal R&D Project

Technologies: Golang, Redis, RabbitMQ, Docker

[GitHub](#)

- Implemented and benchmarked multiple distributed queue technologies to evaluate and compare their performance characteristics.
- Simulated high-concurrency scenarios with workloads ranging from 10 to 100,000 requests, analyzing throughput and latency metrics across repeated benchmark runs.

Achievements

- President Awarded Scout (2018)
- Hackathon - XTREME 2023
- Code Rush 2023 - INTECS UoM (15th/120)
- MoraXtreme 8.0 - IEEEESB UoM

Extra-Curricular Activities

- **AIESEC in Sri Lanka** (2022 – 2024)
 - Organizing Committee Member - External Relations, Lead CS 9.0 (Nov 2022 – Feb 2023)
 - OC Vice President - External Relations, Beauty Beyond Ceylon (May 2022 – Aug 2022)
- Scout Organization - J/Mahajana College (2009–2019)
- Chess - J/Mahajana College (2013–2019)

Referees

Lahiru Madushanka

Associate Architect, Digital Mobility Solution Lanka PLC (Pickme)

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Dr. A.L.A.R.R. Thanuja

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